

Lecture 11: Generative AI

HUM 4441: Engineering Ethics for Computer Science Engineers

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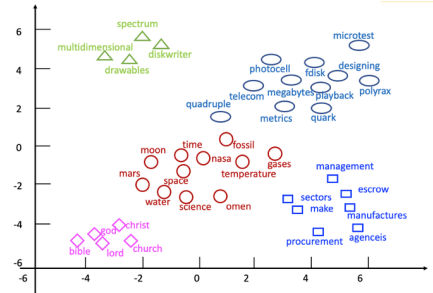
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Language Models: Spatial Word Embedding

- Word = point in higher-dimensional space.
 - Each coordinate is # of occurrences in a particular document.
 - “Nearby” words tend to occur together.
- Dimension reduced from millions to hundreds.
 - Using technique similar to singular value decomposition.



Two-dimensional projection of word embeddings

Recurrent NN Translation

A recurrent NN matches words in one language with words in another that sit in similar spatial positions, taking into account context and word order.

Encoder and Decoder

The network reads the whole source sentence before producing anything, compresses it into one set of numbers, and then generates the target sentence one word at a time from that summary.

the green witch arrived → [numbers] → *llegó la bruja verde*

The adjective ends up after the noun because the summary is translated, once the whole sentence has been read.

Generative Adversarial Networks (GANs)

- Used mainly for images.
- Example: "Draw a cat."
 - **Generator** learns to create a fake image that the **discriminator** will classify as a cat.
 - Discriminator learns to recognize fake images.
 - This adversarial relationship generates realistic but fictitious content.

Generator and Discriminator

The generator starts from random numbers and produces a candidate. The discriminator sees real photos and candidates mixed together and guesses which is which. Each round, both are updated on the result.

When Training Ends

Training stops when the discriminator does no better than chance, which means the fakes now fool a network built to catch them.

Four Uses of the Same Generator

Sort these into acceptable, acceptable with conditions, and never.

1 Synthetic training faces

A team generates 50,000 fake faces to train a face detector, so no real person's photos are collected.

2 A de-aged actor

A studio makes a 70-year-old actor look 30, with the actor's written consent and a fee.

3 Political satire

A student posts a video of a prime minister singing a song he never sang. It is captioned as satire.

4 Your teacher's face

Someone puts your teacher's face into a video they never appeared in, and shares it in a group chat.

GPT = Generative Pre-trained Transformer

- **Generative:** it writes the answer one word at a time instead of retrieving a stored one.
- **Pre-trained:** the training finished before you typed anything. It is not learning from your conversation.
- **Transformer:** the architecture. Attention lets every word be read against every other word in the prompt.

How It Was Trained

By predicting the next word in ordinary text, billions of times over.

The ELIZA Effect

Weizenbaum's own secretary asked him to leave the room so she could talk to it privately. Fluency alone was enough to produce that reaction in 1966, before any of the machinery in this lecture existed.

ChatGPT vs. Traditional Chatbots

- Chatbots date back to the 1960s (e.g., ELIZA)
- ChatGPT isn't really a chatbot.
 - Unclear how to train a chatbot from online data.
- It is very adept at finding text, images, and voice that respond to a prompt.
 - Due to co-occurrence of this material with verbiage similar to the prompt.
 - Based on vast training data and > 1 trillion parameters.

ELIZA, 1966

Joseph Weizenbaum's program imitated a therapist using a few dozen pattern-matching rules — mostly reflecting the user's own sentence back as a question.

*"I am unhappy." →
"Do you think coming here will help
you not to be unhappy?"*

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Two Key Facts about GPTs

Fact 1: They don't know what they are talking about.

Language models don't understand language. They would produce the same output if trained on gibberish (whether or not it encodes meaningful text).

Fact 2: They parrot what has already been said.

With some modifications, cutting and pasting.
Limited ability to distinguish true from false.

When is it ethical to generate documents with GPTs? We look at two arguments; IP rights are covered in the next lecture.

Generalization Argument

Ask what happens if **everyone** did this. A practice that only works because most people don't is not generalizable.

Utilitarian Argument

Weigh the practice's **overall benefit and harm** to society against the alternative.

Note

The industry now euphemistically speaks of *document processing* rather than document generation.

Generalization Argument 1: Implied Promise

An Essay or Article Carries an Implied Promise

- The author has made a good faith effort to say something that is **true**.
- The author has thought about the topic and describes their own understanding or argument.
- In other words, the author is, in fact, the **author**.

GPT Breaks the Promise

Generating a document with a GPT breaks this promise. When done for convenience or profit, this is **not generalizable**.

Generalization Argument 2: The Garbage Loop

The Cycle

- Meaningful material exists for training only because humans have created and researched their own material.
- GPT is already consuming its own output.
- If everyone were to use GPTs... because they create usable material and are convenient... then GPTs would no longer create usable material.
- This is already beginning to occur.



Generalization Argument 2: Exceptions

Routine and Boilerplate Documents

May be generalizable, as they are mainly based on previous work anyway.

Using a GPT as an Assistant

May be generalizable if:

- Fact claims are checked and references given.
- The essay is reorganized and heavily edited to represent the author's own thoughts.



BOILERPLATE

Utilitarian Argument

- By laboring to create one's own documents, one hones research and reasoning skills — which are in short supply.
- One thereby makes a greater positive contribution.
- Reliance on GPTs is therefore **not utilitarian**.
- This applies particularly to students.
- In fact, one can argue that conscientious writing creates literacy.

The Case for Using Them

More output, faster, for everyone. If total benefit is what counts, why is a well-written generated report worse than a badly written honest one?

What It Costs Later

The document is finished once. The writing practice you skipped is gone for good, and the stock of human-written material the next model needs keeps thinning.

When the Code Is Generated

What the Assistant Is Doing

A coding assistant predicts the code that usually follows this code. It has no picture of your threat model or your users, so code that compiles and passes your tests can still be wrong in ways your tests never ask about.

The Copilot Study

In an early study of Copilot on security-relevant tasks, roughly **40%** of generated programs contained a known vulnerability class.

Who Is Accountable

- Professional codes bind whoever **signs** the work.
- Competence means you can account for what you shipped, line by line.
- Where the training data came from is a live legal question (next lecture).

Your generated module passes every test. You cannot explain one function in it. **Ship it or not?**

When Do You Have to Say You Used It?

The Rule

Disclose when the reader's judgment of the work depends on believing that a person wrote it.

Also Disclose When

- A policy, license, or instructor requires it.
- The output carries factual claims you have not personally checked.

Situation	Disclose?
Routine scheduling email	No
Lab report or essay	Yes
Method section of a paper	Yes
Code in a shipped product	Check policy
A draft you rewrote entirely	Judgment call

How much rewriting makes you the author again?

What is Jailbreaking?

Attempts to manipulate an LLM into ignoring or bypassing its safety rules, allowing responses that the model would normally refuse.

Common Techniques

- Role-playing (“Pretend you are. . .”)
- Prompt injection
- Multi-step reasoning attacks
- Obfuscated or encoded prompts
- Context manipulation

Ethical Issue

Using jailbreaks to obtain harmful, illegal, or deceptive information undermines AI safety and responsible use.

Ethics of Jailbreaking

Potential Harm

- Spreading misinformation
- Facilitating cybercrime
- Producing malicious code
- Creating phishing campaigns
- Bypassing safety mechanisms

Responsible Research

- Security researchers test models to improve defenses.
- Findings should be responsibly disclosed.
- The goal is stronger AI safety—not exploiting weaknesses.

Engineering Ethics

Engineers should design AI systems that are robust against misuse while using them responsibly and transparently.

Two students find the same jailbreak. One posts it publicly; the other reports it to the vendor and waits six weeks. Defend the one you would do.

Summary

How It Works

- Words as points in space; nearby words co-occur
- Transformers predict the next word; GANs make synthetic images
- Pre-trained, not reasoning on the fly

Why It Fails

- It does not know what it is talking about
- It parrots, with limited ability to sort true from false
- Hallucinated citations, resumes, policies, code

The Author's Duty

- Authorship carries an implied promise
- Generalization: universal use destroys the input
- Utility: the practice is worth more than the finished document

The lawyer's tool worked as designed. **Every duty in this lecture belonged to a person**, and every failure was a check that somebody skipped.